

Занятие 8

Двойственный симплекс-метод

Решить двойственным симплекс-методом.

1.
$$\begin{aligned} Z &= x_1 + 7x_2 \rightarrow \min \\ \begin{cases} 4x_1 + 7x_2 \geq 57 \\ 4x_1 + x_2 \geq 15 \\ 3x_1 + 11x_2 \geq 60 \\ x_i \geq 0, i=1,2 \end{cases} \end{aligned}$$

2.
$$\begin{aligned} Z &= 6x_1 + x_2 + 4x_3 \rightarrow \min \\ \begin{cases} 2x_1 + x_2 + 5x_3 \geq 12 \\ x_1 + 8x_3 \geq 16 \\ 5x_1 + 2x_2 + x_3 \geq 10 \\ x_i \geq 0, i=1,2,3 \end{cases} \end{aligned}$$

3.
$$\begin{aligned} Z &= 5x_1 + 2x_2 + 9x_3 \rightarrow \min \\ \begin{cases} x_1 + x_2 + x_3 \geq 10 \\ 2x_1 + 3x_2 + 4x_3 \geq 20 \\ 5x_2 + 6x_3 \geq 30 \\ x_i \geq 0, i=1,2,3 \end{cases} \end{aligned}$$

4.
$$\begin{aligned} Z &= x_1 + 9x_2 + 10x_3 \rightarrow \min \\ \begin{cases} 2x_1 + 3x_2 + 5x_3 \geq 24 \\ 3x_1 + x_2 + x_3 \geq 12 \\ x_1 + 4x_2 + 5x_3 \geq 20 \\ x_i \geq 0, i=1,2,3 \end{cases} \end{aligned}$$

5.
$$\begin{aligned} Z &= 4x_1 + x_2 + 5x_3 \rightarrow \min \\ \begin{cases} 3x_1 + 6x_2 + x_3 \geq 18 \\ x_1 + 5x_2 + 4x_3 \geq 20 \\ 3x_1 + 2x_2 + 5x_3 \geq 24 \\ x_i \geq 0, i=1,2,3 \end{cases} \end{aligned}$$

6.
$$\begin{aligned} Z &= 7x_1 + 7x_2 + x_3 \rightarrow \min \\ \begin{cases} 2x_1 + 5x_2 + x_3 \geq 15 \\ x_1 + 6x_3 \geq 14 \\ x_1 + 3x_2 + 8x_3 \geq 24 \\ x_i \geq 0, i=1,2,3 \end{cases} \end{aligned}$$

Ответы:

1. $Z_{\min} = Z(20;0) = 20$

2. $Z_{\min} = Z(0;4;2) = 12$

3. $Z_{\min} = Z(0;10;0) = 20$

4. $Z_{\min} = Z(20;0;0) = 20$

5. $Z_{\min} = Z(0;12;0) = 12$

6. $Z_{\min} = Z(0;0;15) = 15$